Gaspar Zimbrek-Turkovic

Materials engineer, polymers and sustainable thermosets · Lausanne, Switzerland
gasparzimbrek-arch.github.io · LinkedIn · ORCID 0009-0003-8212-0304
Open to PhD positions and R&D roles in polymers and composites from 2027.



RESEARCH

Feb – Aug 2026 **Master's thesis**
Empa, Department of Advanced Fibers, Additives and Chemistry Group (Dr. Cédric Hervieu) · St. Gallen
Reprocessable flame-retardant thermosets based on covalent adaptable networks via aza-Michael addition chemistry.
- Synthesized monomers by Grignard reactions, with column and thin-layer chromatography, crystallization, rotary evaporation and devolatilization.
- Confirmed product structures and the aza-Michael chemistry by NMR (including solid-state NMR of the resin), model reactions, DIP-MS and Raman spectroscopy.
- Prepared β-amino phosphine oxide epoxy (APOE) covalent adaptable network resins by thermal curing; demonstrated self-healing and reprocessing by hot pressing.
- Evaluated flame retardancy (UL 94, cone calorimetry, PCFC) and elemental, thermal and mechanical properties.

Feb – Jun 2025 **Master's research project**
Laboratory for Processing of Advanced Composites (Prof. Véronique Michaud), EPFL · Lausanne
Frontal polymerization of vitrimer composites.
- Investigated thiol-acrylate vitrimer composites with transesterification-based bond exchange, made by frontal polymerization, as sustainable alternatives to thermosets.
- Studied how catalyst concentration (DMAP) and PET microparticle content (0–10 vol%) affect thermal stability, mechanical properties and bond-exchange dynamics.
- Demonstrated shape memory, reprocessability and suitability for circular-economy applications.

Sep 2024 – Jan 2025 **Master's research project**
Laboratory for Soft Materials (Prof. Esther Amstad), EPFL · Lausanne
Double-network granular hydrogels for wearable resistive sensors.
- Developed double-network granular hydrogels from PAMPS microgels and PEG-based deep eutectic solvents.
- Evaluated how PEG molecular weight and hydrogen-bond acceptor type affect swelling, resistivity, rheology and direct-ink-writing (DIW) printability.
- Demonstrated shear-thinning behavior, stress recovery and successful DIW 3D printing.

Jan 2022 – May 2024 **Undergraduate research assistant**
Hickey Research Group, Department of Materials Science and Engineering, Penn State · University Park, PA
Shape-memory triblock copolymer hydrogel fibers produced by wet spinning.
- Investigated how processing parameters (concentration, solvent, strain rate, needle gauge, injection speed) affect fiber processing efficiency, structure and mechanical properties.
- Compared solvents (THF vs. DMF) for actuation stroke, actuation rate and work density of fiber actuators.
- Designed mineralization experiments with SOS hydrogel fibers to replicate collagen fibril mineralization; analyzed crystallization of strained vs. non-strained fibers.
- Wrote MATLAB code to process water-actuation video recordings and quantify actuation kinetics.

SELECTED PROJECT

2025 **OptiFlow, with Decathlon · Runner-up, 2025 EPFL IEE Prize**
MGT-555 Innovation & Entrepreneurship in Engineering, EPFL
AI-powered wearable RFID system with ultra-wideband positioning that tracks shelf stock in real time to improve restocking. Team of six with Decathlon Sports Switzerland. Responsible for hardware development and integration, prototype testing, sustainability analysis and the business case.

PUBLICATIONS

2026 **β-Amino phosphine oxide chemistry enables healable, reprocessable, and flame-retardant epoxy networks**
Hervieu, C., **Žimbrek-Turković, G.**, Cirillo, C., Rupper, P., Sekar, A., Jovic, M., Lehner, S., Gaan, S. · Preprint, SSRN · doi:10.2139/ssrn.7655948

2026 **Recyclable flame-retardant epoxy–amino ester covalent adaptable networks enabled by sustainable polyelectrolyte complexes**
Hervieu, C., Cirillo, C., Lehner, S., **Žimbrek-Turković, G.**, Petri, M., Rupper, P., Jović, M., Sekar, A., Grunlan, J. C., Gaan, S. · Preprint, SSRN · doi:10.2139/ssrn.7323542

EDUCATION

2024 – 2027 **MSc in Materials Science and Engineering**
EPFL · Lausanne, Switzerland
Minor in Management of Technology and Entrepreneurship. Expected graduation February 2027.

2020 – 2024 **BSc in Materials Science and Engineering**
The Pennsylvania State University · University Park, PA, USA
Specialized in polymer science. Graduated May 2024, GPA 3.78 / 4.0. Dean's List every semester except Fall 2020.

AWARDS AND SCHOLARSHIPS

2023 **Penn State representative, Undergraduate Student Speaking Contest, MS&T 2023**
Selected by the Department of Materials Science and Engineering; presented research on biocompatible shape-memory triblock copolymer hydrogel fibers.

Penn State **2nd place of 12 teams, Materials Engineering Methodology and Design Grand Challenge**
Ammonia-based fuel cell for renewable energy storage. Featured in a College of Earth and Mineral Sciences video on undergraduate research.

2022 – 2024 **Scholarships, Penn State**
John and Elizabeth Holmes Teas Scholarship (2022–24) · Penn State Metallurgy Alumni Scholarship (2022–24) · MSE Opportunity Hancock Scholarship (2023–24) · Product Evaluation Systems, Inc. Scholarship (2023–24) · Robert G. and Agnes M. Faber Scholarship (2022–23)

SKILLS AND LANGUAGES

Laboratory **Synthesis and characterization**
Organic synthesis, column chromatography, NMR, IR and Raman spectroscopy, GPC · UV and thermal curing, hot pressing · rheology, DMA, DSC, TGA · UL 94, cone calorimetry, PCFC, LOI · tensile and 3-point bending tests · direct ink writing, FDM 3D printing · optical microscopy, XRD, SEM-EDS, infrared thermography

Software **Data and modelling**
MATLAB, R, Origin, Proteus, Autodesk Moldflow, ChemDraw, PyMOL, CrystalMaker, ImageJ, TRIOS, PrusaSlicer, Microsoft Office, Adobe Photoshop

Languages **Croatian (native) · English (C2) · French (B2–C1) · Italian (A2–B1)**